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III Semester M.C.A. Examination, March/April 2021
(CBCS Scheme)
COMPUTER SCIENCE
MCA 307 : Softcore - Quantitative Teaching and Research Aptitude

Time : 3 Hours Max. Marks : 70

Instructions : 1) Answer ***any five*** from Part – ***A***.
2) Answer ***any four*** from Part – ***B***.

PART – A

Answer **any five** of the following. **Each** question carries **six** marks : **(5×6=30)**

1. In a mixture of 60 litres, the ratio of milk and water is 2 : 1. If this ratio is to be 1 : 2, then what quantity of water to be added ?

2. Raju, Kushal and Virat start at same time, same point and in same direction to run around a circular ground. Raju completes a round in 250 seconds, Kushal in 300 seconds and Virat in 150 seconds. Find after what time will they meet again at the starting point ?

3. Naveen purchased 120 reams of paper at Rs.80 per ream. He spent Rs. 280 on transaction, paid octroi at the rate of 40 paisa per ream and paid Rs. 72 to the servant. If he wants to have a gain of 8%, what must be the selling price per ream ?

4. The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 3 years old. If ages of these two are excluded, the average age of remaining players is one year less than the average age of the whole team. What is the average age of the team ?

5. Rohan took a loan of Rs. 1,500 with simple interest for as many years as the rate of interest. If he paid Rs. 540 as interest at the end of loan period, what was the rate of interest ?

6. Briefly explain the elements involved in the teaching process.

7. List and explain the characteristics of good listener as well as good teacher.

8. Distinguish between Seminar, Conference and Symposium.

P.T.O.

PART – B

Answer **any four** of the following. **Each** question carries **10** marks : **(4×10=40)**

9. a) Give a brief introduction about regulatory framework of higher education in India. 5

   b) Write a note on Online Education System. Will it be a boon or bane ? Justify your answer. 5

10. a) After two successive discounts, a tie with a list price of Rs. 120 is available at Rs. 90. If second discount is 9 %, what is the first discount ? 5

    b) A man can row 40 km upstream and 55 km downstream in 13 hours. Also, he can row 30 km upstream and 44 km downstream in 10 hours. Find the speed of the man in still water and the speed of the current. 5

11. a) The distance between two cities Chennai and Bengaluru is 330 km. A train starts from Chennai at 8 a.m. and travels towards Bengaluru at 60 kmph. Another train starts from Bengaluru at 9 a.m. and travels towards Chennai at 75 kmph. Find at what time do they meet ? 5

    b) 10 women can complete a work in 7 days and 10 children take 14 days to complete the same work. How many days will 5 women and 14 children take to complete the work ? 5

12. a) One year ago, the ratio of Manoj and Sachin's age was 6 : 7 respectively. Four years hence, this ratio would become 7 : 8. How old is Sachin ? 5

    b) The price of a car is Rs. 3,25,000. It was insured to 85% of its price. The car was damaged completely in an accident and insurance company paid 90% of the insurance. What was the difference between the price of car and amount received ? 5

13. a) A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/h, find the length of the platform ? 5

    b) The top of a 15m high tower makes an angle of elevation of 60 degrees with the bottom of an electric pole and angle of elevation of 30 degrees with the top of the pole. What is the height of the electric pole ? 5

14. a) A bag contains 6 Red and 4 Blue balls. Two balls are drawn at randomly. Find the probability that they are of the same color. 5

    b) What was the day of the week on 12th January 1863 ? 5

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III Semester M.C.A. Degree Examination, April/May 2022
(CBCS Y2K20 Scheme)
COMPUTER SCIENCE
3MCA2 : Quantitative, Teaching and Research Aptitude (Soft Core)

Time : 3 Hours Max. Marks : 70

Instructions : *Answer **any five** from Part – **A** and answer **any four** from Part – **B**.*

PART – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Of the three numbers, second is twice the first and is also thrice the third. If the average of the three numbers is 55. Find the largest number.
2. Simplify.

$$\frac{1}{\log_4 120}+\frac{1}{\log_5 120}+\frac{1}{\log_6 120}$$

3. Two pipes A and B can fill a tank in 10 hours and 12 hours respectively while the third pipe empties the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time will the tank be filled ?
4. At what time between 4 and 5 O'clock will the hands of a clock be at right angle ?
5. The average of five consecutive numbers A, B, C, D and E is 48. What is the product of A and E ?
6. The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher ?
7. Explain workshop, conference and symposium.
8. Explain governance, policy and administration of higher education system in India.

PART – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) 2 men and 3 boys can do a piece of work in 10 days while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy can do the work ? **5**

P.T.O.

b) A train starts full of passengers. At the first station, it drops one-third of the passengers and takes 280 more. At the second station, it drops one-half of the new total and takes 12 more. On arriving at the third station, it is found to have 248 passengers. Find the number of passengers in the beginning. **5**

10. a) A dealer sold three-fourth of his articles at a gain of 40% and the remaining at cost price. Find the gain earned by him in the whole transaction. **5**

b) John's age after six years will be three-seventh of his father's age. Ten years ago, the ratio of their ages was 1:5. What is John's age at present ? **5**

11. a) On dividing a certain number by 342, we get 47 as remainder. If the same number is divided by 18, what will be the remainder ? **5**

b) The difference between the simple interest and true discount on a certain sum of money for 6 months at 121/2% per annum is Rs. 25. Find the sum. **5**

12. a) Define teaching. Explain the nature of teaching aptitude. **5**

b) Explain the factors affecting the teaching. **5**

13. Explain the steps, methods and ethics of research. **10**

14. a) Explain the characteristics and types of communication. **5**

b) Difference between formal education and distance education in India. **5**

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III Semester M.C.A. (Two Years Course) Examination, May/June 2023
(CBCS) (2021 – 22)
COMPUTER SCIENCE
3MCA2 : Quantitative, Teaching and Research Aptitude (Soft Core)

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Part – **A** and answer **any four** from Part – **B**.*

PART – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44. Find the largest numbers.
2. What will be the possibility of drawing a Jack or a Spade from a well shuffled standard deck of 52 playing cards ?
3. Simplify $\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60}$.
4. Which day of the week was 18th April 1963 ?
5. At what time between 3'O clock and 4'O clock the hands of the clock will be together ?
6. Two pipe A and B can fill a tank in 4 hrs. and 5 hrs. respectively. If they opened on alternate hours and if pipe B is opened first. In how many hours the tank is full ?
7. Define the term teaching. Explain the nature and objectives of teaching in brief.
8. Briefly discuss the research methodologies and its types.

PART – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) In how many ways can we arrange the word "FUZZTONE" so that all the vowels come together ? **5**

   b) 50 Labourers can dig a pond in 16 days. How many labourers will be required to dig another pond in 20 days, which is double in size ? **5**

P.T.O.

10. a) 12 years ago, age of P was 3 times the age of Q. After 12 years, ratio of ages to Q to P will be 2 : 3. What is the present age of P ? 5

    b) At what dates of April 2001 did wednesday fall ? 5

11. a) Explain the characteristic and types of communication. 5

    b) Difference between formal education and distance education in India. 5

12. a) Explain governance policy and administration of higher education system in India. 5

    b) Define the following terms. 5

    i) Article

    ii) Workshop.

13. a) A can do $\frac{1}{2}$ of a piece of work in 5 days, B can do $\frac{3}{5}$ of the same work in 9 days and C can do $\frac{2}{3}$ of that work in 8 days. In how many days can all of them together do the work ? 5

    b) At what time between 7'O clock and 8'O clock the hands of the clock be in opposite direction. 5

14. a) The compound interest on Rs. 30,000 at 7% per annum for the certain time is Rs. 4,347. Find the time taken in years. 5

    b) Ajay started a business investing Rs. 12,000. After 5 months, Bala joined him with a capital of Rs. 9,000. At the end of the year the total profit was Rs. 4,416. What is the share of profit of Ajay ? 5

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III Semester M.C.A. (Two Years Course) Degree Examination, May 2024
(CBCS) (2021 – 22)
COMPUTER SCIENCE
3MCA2 : Quantitative, Teaching and Research Aptitude

Time : 3 Hours Max. Marks : 70

***Instruction** : Answer **any five** questions from Section – **A** and **any four** questions from Section – **B**.*

SECTION – A

Answer **any five** questions. **Each** question carries **six** marks. **(5×6=30)**

1. Find the H.C.F. and L.C.M. of 2/3, 8/9,16/81 and 10/27.

2. The age of father 10 years ago was thrice the age of his son. 10 years hence father's age will be twice that of his son. Find the ratio of their present ages.

3. A Man spends 2/5 of his salary on house rent, 3/10 of his salary on food and $\frac{1}{8}$ of his salary on conveyance if he has $1400 left with him, find his expenditure on food and conveyance.

4. Divide $1050 among A, B and C such that A receives 2/5 as much as B and C together and B receives 3/7 as much as A and C together.

5. At what time between 4 and 5 o'clock will the hands of a clock be at right angle ?

6. Explain the learner's characteristics in teaching.

7. A man divides 8,600 among 5 sons, 4 daughters and 2 nephews. If each daughter receives five times as much as each nephew and each son receives four times as much as nephew, how much does each daughter receive ?

8. Define Research. Explain the types of research in details.

SECTION – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) A, B and C enter into a partnership with capitals in the ratio 7/2 : 4/3 : 6/5. After 4 months A increases his share of capital by 50%. If at the end of the year the total profit earned is $2430, find the share of each in the profit.

   b) How many arrangements can be made out of the letters of the word 'ENGINEERING' ?

10. a) A can complete a work in 10 days, B in 12 days and C in 15 days. All of them began the work together, but A had to leave the work after 2 days of the start and B, 3 days before the completion of the work. How long did the work last ?

    b) A man on the top of a tower, standing on the seashore, finds that a boat coming towards him takes 10 minutes for the angle of depression to change from 30 degree to 60 degree. Find the time taken by the boat to reach the shore form his position.

11. a) A person reaches his destination 40 minutes late if his speed is 3 km/hr, and reaches 30 minutes before time if his speed is 4 km/hr. Find the distance of his destination from his starting point.

    b) A man can row 6 km/hr in still water. It takes him twice as long to row-up as to row-down the river. Find the rate of stream.

12. a) Briefly explain the factors affecting the teaching.

    b) Define communication. Explain Barriers in effective Communication.

13. a) Prove that the calendar for the year 2003 will serve for the year 2014.

    b) The minute hand of a clock overtakes the hour hand at intervals of 65 minutes of the correct time. How much a day does the clock gain or lose ?

14. a) Define Thesis Writing, its characteristics and format.

    b) Write a short note on :

       i) Formal and Distance Education.

       ii) Value Education.

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III Semester M.C.A. (Two Years Course) Examination, May/June 2025
(CBCS) (2021-22)
COMPUTER SCIENCE
3MCA2 : Softcore : Quantitative, Teaching and Research Aptitude

Time : 3 Hours Max. Marks : 70

Instructions : *1) Answer **any five** from Section – **A**.*
*2) Answer **any four** from Section – **B**.*

SECTION – A **(5×6=30)**

1. A television manufacturer earns 20% profits by selling each T.V. set for Rs. 14,400. If the production cost is increased by 15% what should be the new selling price of a set so as to gain 15% ?

2. A person A had 85 currency notes in all, some of which were of Rs. 100 denomination and the remaining of Rs. 50 denomination. The total amount of all these currency notes was Rs. 5,000. How much did he have in the denomination of Rs. 50 ?

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3. What was the day of the week on 26th January, 1950 ?

4. A tap can fill a tank in 10 minutes and another can empty it in 6 minutes. If the tank is already two-fifths full can both the taps are opened together, will the tank be filled or emptied ? How long will it take before the tank is either filled completely or emptied completely, as the case may be ?

5. The product of the ages of Ankit and Nikita is 240. If twice the age of Nikita is more than Ankit's age by 4 years, then find Nikita's age.

6. A bag cntains 6 white and 4 black balls. Two balls are drawn at random. Find the probability that they are of same color.

7. Explain the objective, characteristics of learners and teachers.

8. Explain value education.

P.T.O.

SECTION – B (4×10=40)

9. a) If 30% of (P+Q) = 50% of (P–Q), then find P:Q.

   b) In how many ways can the letters of the word ARITHMETIC be arranged so that, the vowels never appear together ?

10. a) A goods train leaves a station at a certain time and a fixed speed. After 6 hours, an express train leaves the same station and moves in the same direction at a uniform speed of 90 kmph. The train catches up the goods train in 4 hours. Find the speed of the goods train.

    b) The angle of elevation of the top of a tower at a point on the ground is 30 degree. On walking 24 m towards the tower, the angle of elevation becomes 60 degree. Find the height of the tower.

11. a) At what time between 4 and 5 o'clock will the hands of a clock be at right angle ?

    b) How many kgs. of wheat costing Rs. 8 per kg must be mixed with 36 kg of rice costing Rs. 5.40 per kg so that 20% gain may be obtained by selling the mixture at Rs. 7.20 per kg ?

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12. a) Briefly explain the factors affecting the teaching methodologies.

    b) What are the steps involved in research ?

13. a) Explain the structure of the institution for higher education and research in India.

    b) Highlight the key roles of department of higher education in India.

14. Write a short note on :

    a) Communication

    b) Formal and distance education

    c) Professional and technical education.

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III Semester M.C.A. (Two Years Course) Examination, May/June 2026
(CBCS Scheme) (2021-22)
COMPUTER SCIENCE
3MCA2 : Quantitative, Teaching and Research Aptitude

Time : 3 Hours Max. Marks : 70

Instructions : 1) *Answer **any five** questions from Section – **A**.*
2) *Answer **any four** questions from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. a) Find the H.C.F. and L.C.M. of $\frac{2}{3}, \frac{8}{9}, \frac{16}{81}$ and $\frac{10}{27}$.

   b) Evaluate

   i) $\frac{35}{0.7}$ ii) $\frac{2.5}{0.0005}$ iii) $\frac{136.09}{43.9}$

2. The average weight of A, B and C is 45Kg. If the average weight of A and B is 40 kg and that of B and C is 43 Kg, find the weight of B.

3. A book was sold for Rs. 27.50 with a profit of 10% if it were sold for Rs. 25.75, then what would have been the percentage of profit or loss ?

4. A, B and C start a business, each investing Rs. 20,000. After 5 months, A withdraws Rs. 5,000, B withdraw Rs. 4,000 and C invests Rs. 6,000 more. At the end of the year, a total profit of Rs. 69,900 was recorded. Find the share of each.

5. A and B can do a piece of work in 18 days, B and C can do it in 24 days, and A and C can do it in 36 days. In how many days will A, B and C finish it working together and separately ?

6. A bag contains 6 white and 4 green balls. Two balls are drawn at random. Find the probability that there are of the same colour.

7. Discuss softcore and hardcore briefly.

8. Define communication. Explain the barriers to effective communication.

P.T.O.

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SECTION – B

Answer **any four** of the following. **Each** question carries **ten** marks. **(4×10=40)**

9. a) The sum of the squares of three consecutive odd numbers is 2531. Find the numbers.

   b) A thief is spotted by a policeman from a distance of 100 meters. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8 km/hr. and that of the policemen is 10 km/hr., how far will the thief have run before he is overtaken ?

10. a) How many words can be formed from the letters of the word 'DIRECTOR' so that the vowels are always together ?

    b) At what time, between 5 and 6 o'clock, are the hands of the clock 3 minutes apart ?

11. a) Two pipes can fill a cistern in 14 hours and 16 hours, respectively. The pipes are opened simultaneously, and it is found that due to the leakage in the bottom, it took 32 minutes more to fill the cistern. When the cistern is full, how long will it take for the leak to empty it ?

    b) Find the day of the date 26th August 1950 ?

12. a) A man can row 18 km/hr. in still water. It takes him thrice as long to row up as to row down the river. Find the rate of stream.

    b) A ladder leaning against the wall makes an angle of 60° with the ground. If the length of the ladder is 19 meters, find the distance of the foot of the ladder from the wall.

13. a) Write a note on :
    i) Paper and Article
    ii) Workshop and Seminar
    iii) Conference and Symposium.

    b) Explain the factors affecting teaching with respect to teaching methods.

14. a) Explain the various steps of the research process with the diagram.

    b) Explain value education in detail.

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